

Continuous distributions and densities

Probability is area under a density curve, and a few named curves cover most data.

THE QUESTION

How do we give chances to a quantity that can take any value in a range, and which curve fits which data?

THE INTUITION

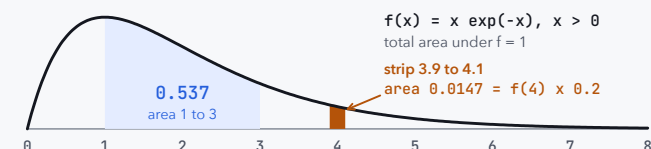
A density $f(x)$ is probability per unit of x : the chance of an interval is the area under f over it. A single point has no width, so it has no chance.

KEY TAKEAWAY

$P(a < X \leq b) = \int_a^b f(x) dx = F(b) - F(a)$. A density needs $f \geq 0$ and area 1, and can exceed 1. Pick a family by the support.

density $f(x)$ → area over $(a, b]$ → $= P(a < X \leq b)$ → running area: CDF $F(x)$ → slope of F : $f(x)$

1 Probability is area under the density



A continuous X has a density $f(x) \geq 0$ with $P(a < X \leq b) = \int_a^b f(x) dx$. For $f(x) = xe^{-x}$ on $x > 0$, $P(1 < X \leq 3) = 0.537$. A thin strip of width Δ holds about $f(x)\Delta$, so f is probability per unit of x . Shrink the strip to one point and its area goes to 0: $P(X = x) = 0$ for every x , so $<$ and $\leq$ give the same answer. Libraries return the log density: in PyTorch, `dist.log_prob(x).exp()` gives $f(x)$.

2 A density can be larger than 1



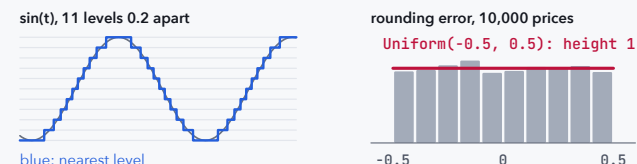
A valid PDF needs only $f(x) \geq 0$ and $\int f(x) dx = 1$. Uniform(0, 2) has height 0.5; squeeze it onto (0, 0.5) and the height must be 2 to keep the area at 1. Exponential(5), mean 0.2, starts at $f(0) = 5$. A height is not a probability; height times a small width is. A mix $wf_1 + (1-w)f_2$ with $0 \leq w \leq 1$ is again a valid PDF.

3 The CDF is the running area; its slope is f



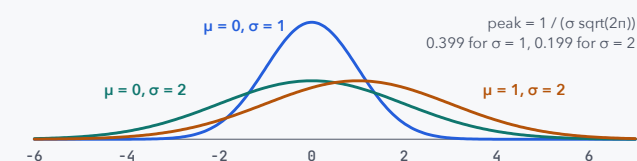
$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$ and $f(x) = F'(x)$. For $f(x) = xe^{-x}$, $F(x) = 1 - (1+x)e^{-x}$: $F(2) = 0.594$ and $P(1 < X \leq 3) = F(3) - F(1) = 0.801 - 0.264 = 0.537$. F climbs fastest where f peaks: slope $f(1) = 0.368$ at $x = 1$. Quantiles and inverse-CDF sampling work as on the CDFs and sampling sheet.

4 Uniform: flat over a range



Uniform(a, b) has $f(x) = 1/(b-a)$ on $[a, b]$ and 0 outside. Rounding prices to the nearest rupee leaves an error close to Uniform(-0.5, 0.5) (10,000 illustrative prices, seed 0). Its size is 0.25 on average and above 0.4 for 20% of prices (simulated: 0.249, 19.1%). Quantising $\sin(t)$ to levels 0.2 apart leaves errors within ± 0.1 , modelled as Uniform(-0.1, 0.1) of height 5.

5 Normal: centre μ , spread σ



$f(x) = 1/(\sigma\sqrt{2\pi}) \cdot \exp(-(x-\mu)^2/(2\sigma^2))$ on all reals. It is symmetric about μ , so mean, median and mode all equal μ . Doubling σ doubles the width and halves the peak. Notation trap: maths writes $\mathcal{N}(\mu, \sigma^2)$ with the variance, but `torch.distributions.Normal(mu, sigma)` and `rng.normal(mu, sigma)` take σ .

6 68, 95 and 99.7 percent within 1, 2, 3 σ



The standard normal CDF is $\Phi(z) = (1 + \text{erf}(z/\sqrt{2}))/2$. $P(|Z| \leq 1) = 0.6827$, $P(|Z| \leq 2) = 0.9545$, $P(|Z| \leq 3) = 0.9973$; each tail is half the rest. Exactly 95% lies within ± 1.96 . Every normal's CDF is 0.5 at its mean.

```
d = torch.distributions.Normal(0., 1.)
z = torch.tensor([0., 1.96])
d.log_prob(z).exp() # 0.3989, 0.0584
d.cdf(z) # 0.5000, 0.9750
```

Standard scores, waits and other shapes

Rescale any normal to one standard curve, model waits with the exponential, and match shapes to the data's range.

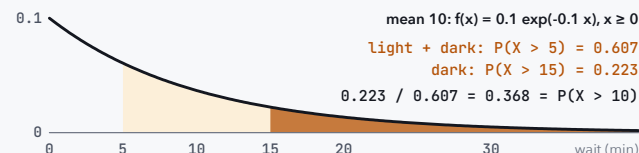
7 Standardise to read any normal

mean 70, $\sigma = 8$
 $z = (x - 70) / 8$



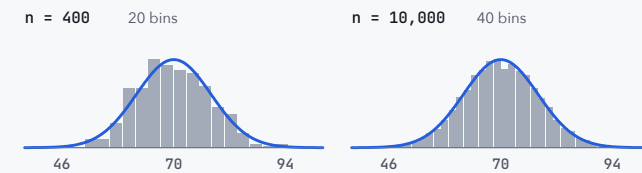
$Z = (X - \mu) / \sigma$ is standard normal, so $P(X \leq x) = \Phi((x - \mu) / \sigma)$ for every normal. Toy model: marks with mean 70 and $\sigma = 8$ (the source notebook itself warns against grading on a bell curve). 86 sits at $z = 2$, so $P(X > 86) = 1 - \Phi(2) = 2.28\%$; below 58 ($z = -1.5$) is 6.68%. Backwards, the 90th percentile is $70 + 1.2816 \times 8 = 80.3$ (icdf(0.9) in PyTorch).

8 Exponential waits forget the past



$f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$, mean $1/\lambda$: the continuous cousin of the geometric. Waits averaging 10 minutes have $\lambda = 0.1$ and $P(X > x) = e^{-0.1x}$, so $P(X > 10) = e^{-1} = 0.368$. Memoryless: after 5 minutes, the chance of 10 more is $P(X > 15) / P(X > 5) = 0.368$, as for a fresh wait. The CDF, the median 6.9 and sampling by $-\ln(1 - u) / \lambda$ are on the CDFs and sampling sheet.

9 Density histograms sit on the scale of f



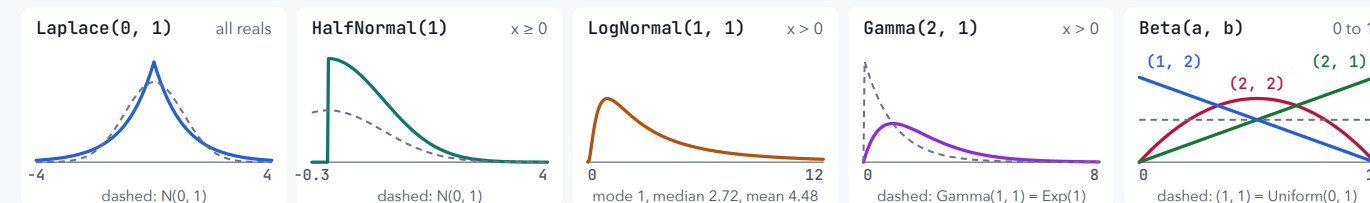
density=True divides each count by $n \times \text{bin width}$, so the bar areas add to 1 and the heights estimate f (blue). 400 marks, mean 70, $\sigma = 8$, seed 0: the bin 69.9 to 72.7 holds 48, height $48 / (400 \times 2.74) = 0.044$, near $f(70) = 0.050$. The largest gap at bin centres is 0.010; with 10,000 draws, 0.005.

```
rng = np.random.default_rng(0)
x = rng.normal(70, 8, 400)
h, e = np.histogram(x, 20, density=True)
(h * np.diff(e)).sum() # 1.0
```

10 Pick a family by where the data can live

First ask which values are possible. **All reals** (errors, noise): normal, or Laplace when big deviations are common. **Positive** (waits, sizes): exponential, gamma, half-normal, or log-normal when $\ln x$ looks bell-shaped. **0 to 1** (chances, shares): beta or uniform. A normal for waits puts some chance on negative waits. `dist.support` gives the allowed range, and PyTorch's `log_prob` raises an error outside it, so mask such points to density 0.

11 Five more shapes, each with its own support



Laplace(μ, b): $f = e^{-|x - \mu|/b} / (2b)$. Heavier tails than $\mathcal{N}(0, 1)$: $P(|X| > 3) = 0.050$ against 0.0027.
Gamma(k, θ): $f = x^{k-1} e^{-x/\theta} / (\Gamma(k) \theta^k)$, mean $k\theta$; $k = 1$ is the exponential. PyTorch's Gamma takes the rate $1/\theta$.

Half-normal(σ): $|Y|$ for $Y \sim \mathcal{N}(0, \sigma^2)$, twice the normal's height on $x \geq 0$; mean 0.798σ .
Beta(α, β): $f \propto x^{\alpha-1} (1-x)^{\beta-1}$ on $[0, 1]$, mean $\alpha / (\alpha + \beta)$; for chances and shares.

Log-normal(μ, σ): $\ln X \sim \mathcal{N}(\mu, \sigma^2)$. For (1, 1): mode 1, median 2.72, mean $e^{1.5} = 4.48$.
Fitting: pick the parameters that make the data most likely, minimising `-d.log_prob(x).mean()` (the likelihood sheet).

12 Pause and think

Can a density equal 3? Yes: Uniform(0, 1/3) has height 3.
Marks, mean 70, $\sigma = 8$. $P(X = 70)$? 0; $P(69.5 < X \leq 70.5) = 0.050$.

$f(x) = c$ on $[0, 4]$. What is c ? $1/4$, so that the area $4c$ is 1.
Share of marks below 54? 2.28%, the mirror image of above 86.

$f(x) = 2x$ on $[0, 1]$: valid? $P(X \leq 0.5)$? Yes, area $1 \cdot 0.5^2 = 0.25$.
Waited 20 min (mean 10). Chance of 10 more? $e^{-1} = 0.368$ again.

$P(a < X \leq b) = \text{area under } f \cdot P(X = x) = 0 \cdot f \geq 0$, area 1, may exceed 1 · $F' = f \cdot Z = (X - \mu) / \sigma \cdot \text{family by support}$

Source: the course notebook "PDF of some common continuous random variables and applications" (normal, Laplace, half-normal, log-normal, gamma, uniform and quantisation error, beta, exponential, with the parameters drawn here and 400 marks with mean 70 and $\sigma = 8$) and the continuous distributions app, nipunbatra.github.io/psdv-teaching. Rounding prices, waits and the other examples are our own; all numbers are computed by the sheet's figure script with seed 0. Next: the expectation and variance sheet.